MULTI-PLAN TELECOMMUNICATIONS COST OPTIMIZATION

1. Problem Statement

Evaluate multiple mobile plans based on a user's monthly data usage, call minutes, SMS count, validity requirement, and budget. Calculate the total cost of each plan, including excess usage charges. Identify and display the most economical plan that satisfies all the given constraints.

2. Objective

To compare mobile plans, calculate additional charges for usage beyond plan limits, check validity and budget conditions, and select the lowest-cost eligible plan.

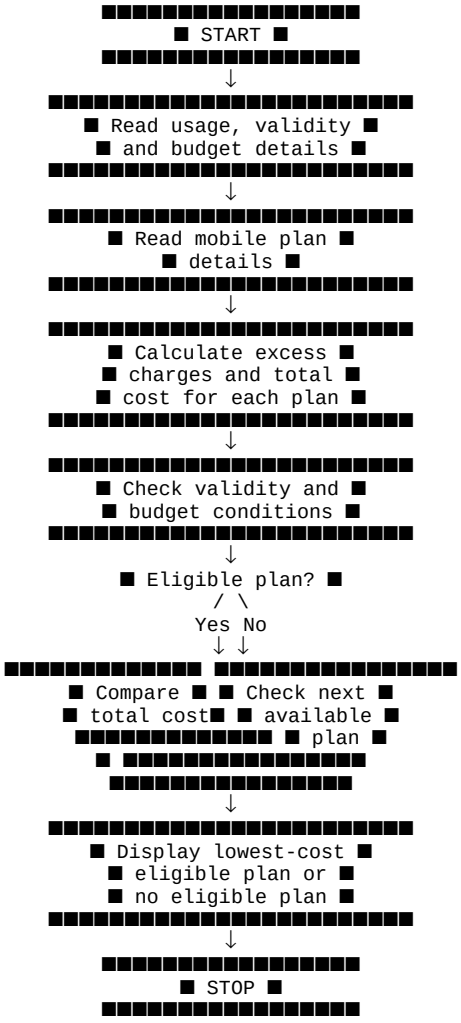
3. Assumptions

1. All plans are evaluated for one billing period.
2. Each plan has a fixed price, data limit, call-minute limit, SMS limit, validity, and excess rates.
3. Excess charges are calculated only for usage above the included limits.
4. A plan is eligible only if its validity meets the required days and its total cost is within the user's budget.
5. If no plan satisfies all conditions, the program displays that no eligible plan is available.

4. Algorithm

1. Start.
2. Read the user's data usage, call minutes, SMS count, required validity, and budget.
3. Store the details of available mobile plans.
4. For each plan, calculate excess data, call, and SMS charges.
5. Add excess charges to the base price to get the total cost.
6. Check whether the plan meets the validity and budget requirements.
7. Among eligible plans, find the plan with the lowest total cost.
8. Display each plan's cost and the most economical eligible plan.
9. If no plan is eligible, display a suitable message.
10. Stop.

5. Flowchart (Vertical)



6. Pseudocode

START
INPUT data_usage, call_usage, sms_usage
INPUT required_validity, budget
plans = list of mobile plans
best_plan = NONE
FOR each plan IN plans
  extra_data = MAX(0, data_usage - plan.data_limit)
  extra_calls = MAX(0, call_usage - plan.call_limit)
  extra_sms = MAX(0, sms_usage - plan.sms_limit)
  extra_cost = extra_data * plan.data_rate
    + extra_calls * plan.call_rate
    + extra_sms * plan.sms_rate
  total_cost = plan.price + extra_cost
  IF plan.validity >= required_validity AND total_cost <= budget THEN
    IF best_plan is NONE OR total_cost < best_plan.cost THEN
      best_plan = current plan and total_cost
    END IF
  END IF
END FOR
IF best_plan is NONE THEN
  DISPLAY "No eligible plan found"
ELSE
  DISPLAY best_plan details and total_cost
END IF
STOP

7. Python Code

```
plans = [
    {"name": "Basic", "price": 199, "data": 2, "calls": 200,
     "sms": 100, "validity": 28, "data_rate": 20,
     "call_rate": 0.5, "sms_rate": 0.2},
    {"name": "Smart", "price": 299, "data": 3, "calls": 500,
     "sms": 200, "validity": 56, "data_rate": 15,
     "call_rate": 0.4, "sms_rate": 0.2},
    {"name": "Premium", "price": 399, "data": 5, "calls": 800,
     "sms": 300, "validity": 84, "data_rate": 10,
     "call_rate": 0.3, "sms_rate": 0.1}
]

data_usage = float(input("Enter data usage in GB: "))
call_usage = int(input("Enter call minutes: "))
sms_usage = int(input("Enter SMS count: "))
required_validity = int(input("Enter required validity in days: "))
budget = float(input("Enter your budget: "))

best_plan = None
best_cost = 0

for plan in plans:
    extra_data = max(0, data_usage - plan["data"])
    extra_calls = max(0, call_usage - plan["calls"])
    extra_sms = max(0, sms_usage - plan["sms"])

    extra_cost = (extra_data * plan["data_rate"] +
                  extra_calls * plan["call_rate"] +
                  extra_sms * plan["sms_rate"])

    total_cost = plan["price"] + extra_cost
    print(plan["name"], "total cost: Rs.", round(total_cost, 2))

    if plan["validity"] >= required_validity and total_cost <= budget:
        if best_plan is None or total_cost < best_cost:
            best_plan = plan["name"]
            best_cost = total_cost

if best_plan is None:
    print("No eligible mobile plan found.")
else:
    print("Most economical eligible plan:", best_plan)
    print("Total cost: Rs.", round(best_cost, 2))
```

8. Expected Output

Enter data usage in GB: 2.5
Enter call minutes: 300

Enter SMS count: 120
Enter required validity in days: 28
Enter your budget: 350

Basic total cost: Rs. 309.0
Smart total cost: Rs. 299.0
Premium total cost: Rs. 399.0
Most economical eligible plan: Smart
Total cost: Rs. 299.0